



Original Research Article

Conversion of waste bamboo chopsticks to bio-oil via catalytic hydrothermal liquefaction using K_2CO_3 

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ABSTRACT

Hydrothermal liquefaction (HTL) was employed to transform biomass into bio-oil. Waste bamboo chopsticks (WBC) were utilized as the raw materials of HTL reaction for resolving excessive WBC problem. Effects of temperatures (T_C) from 563 to 653 K and K_2CO_3 catalyst on the yield (Y_{BOAS}) of bio-oil ($BOAS$) obtained as acetone soluble were assessed. As illustrations, for WBC without catalyst at T_C of 593 K with reaction time of 30 min, the yield Y_{BOAS} reaches only 3.8 wt% with heating value (H_{BOAS}) of 30.9 MJ kg⁻¹, while the addition of catalyst (catalytic HTL, CHTL) with T_C at 563 K can lead to evident enhancement of Y_{BOAS} to 21.2 wt% with H_{BOAS} of 31.6 MJ kg⁻¹. The yield (Y_{BOCC}) of organic C of $BOAS$ with respect to C of dry-ash-free raw WBC for CHTL corresponding to 21.2 wt% of Y_{BOAS} is 33.1 wt%. Higher Y_{BOAS} and carbon content of $BOAS$ give higher Y_{BOCC} and H_{BOAS} , respectively. The results indicate that K_2CO_3 is an effective catalyst for HTL of biomass of WBC with significant lignin content of 25.5 wt% in dry-basis. Furthermore, the $BOAS$ shows simulated distillation characteristic close to diesel, while is superior to heavy oil and boat oil.

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1. Introduction

Since the decrease of fossil fuel, the development of renewable bio-energies has drawn much attention around the world. They rank the 4th energy following oil, coal, and natural gas [1]. Among them, the bio-waste to energy is of special interest [2–10]. One of the problematic wastes in Asia is the waste bamboo chopsticks (WBC) [4,11–15].

Various technologies are available for the conversion of biomass and or bio-waste to different end uses of energy including biochemical, thermal chemical and physical processes [2–11,13,14,16–23]. The energy converted from biomass and bio-

waste could be in solid, liquid, or gas forms [2]. The solid energy is usually the solid bio-fuel with upgraded energy and fuel qualities. The gaseous energy may be produced as bio-gas, hydrogen, or synthetic gas. Comparing with solid and gas fuels, liquid bio-fuels are more advantageous in many aspects such as easy storage and transportation, and diverse uses.

Among the processes producing liquid bio-fuel, the method of hydrothermal liquefaction (HTL) has been applied by many investigators due to the simplicity of the process [9,10,16–25]. Brand et al. [24] indicated that the HTL mechanism involves hydrolysis in 473–563 K, pyrolysis in 563–643 K, and recombination, depending on temperature, heating rate, and residence time. The recombination becomes pronounced with high heating rate and long residence time. The respective temperatures for pyrolysis and recombination decrease with increasing heating rate and residence time. All these mechanisms and affecting factors thus influence the product distribution. Sugano et al. [25] studied product yields after the HTL of

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eucalyptus with paper regeneration wastewater in 423–623 K. The oil yield was about 8 wt% dry ash free (daf) at 423 K, while increased to about 34 wt% at 573 K. It, however, decreased to about 25 wt% at 623 K. Xu and Lancaster [9] investigated the liquefaction of secondary pulp/paper sludge powder in hot-compressed water in 523–653 K for 60 min. The yield of heavy oil was around 15 wt% daf at 523 K, increasing to around 24 wt% at 623 K and then decreasing to around 19 wt% at 653 K. These results indicated that the effect of temperature on the oil yield is not monotonic.

Different catalysts have been added to enhance the HTL reactions [10,17,18,21,23]. Among the catalysts used, alkaline, a common chemical, exhibits enhancement effect on HTL as noted by others [10,23]. This study thus employed K_2CO_3 catalyst in HTL to transform biomass of WBC into bio-oil (BO), which has not been reported before. Effects of key parameters of temperature (T_C), reaction time (t_{rCT}) for constant-temperature reaction period (i.e., at constant T_C), and K_2CO_3 catalyst on the performance of HTL were elucidated. These include the yield (Y_{BOAS}), heating value (H_{BOAS}), and chemical characteristics of BO (BO_{AS}) obtained as acetone soluble.

2. Experimental methods

2.1. Materials and chemicals

The disposable bamboo chopsticks (Songya Co., Taipei, Taiwan) were used to simulate the WBC. They were pulverized with particle size of 0.5–2 mm for HTL reaction. Solvent and catalysts used were Q water (Pure Q Ultra Pure Water System, Systematic Instruments, Singapore) and K_2CO_3 , respectively.

2.2. Operations of HTL and catalytic HTL

WBC (10 g, dry-basis) and DI water (100 mL) filled into reactor (HP/HT pressure reactor, 4563 Bench Top Reactor, Parr Instrument, Moline, IL) have fixed ratio (1:10 in g:mL) in all experimental conditions. Nitrogen (99.995%, Ching-Fong Co., Taipei, Taiwan) was set to remain 345 kPa (gauge reading) for 5 min to remove oxygen in reactor before HTL reaction. After that, rapid heating rate was given. Stirring speed was set at 200 rpm. The tested setting temperatures T_C were 563, 593, 623, and 653 K, respectively, while reaction times t_{rCT} at constant T_C was 30 min. For catalytic HTL (CHTL), 2 g K_2CO_3 were added. The use of 20 wt% of K_2CO_3 relative to WBC was aimed at assessing its catalytic feasibility on the performance of the process. The optimal or suitable amount of catalyst less than 20 wt% would be determined by further tests employing various amounts of catalyst. Of course, less catalyst with acceptable performance would be desirable. The experimental scheme of the above batch process is illustrated in Fig. 1.

For conducting the HTL, the autoclave reactor was heated from 298 K (25 °C) to the desired setting temperature T_C and held for a period of time of t_{rCT} . It was then cooled down to 298 K to end the experiment. It took about 90 min for heating to reach T_C of 653 K (380 °C) and about 95 min for cooling down to 298 K. Fig. 2 illustrates the variation of temperature with overall reaction time (t_{OA}) from the beginning of reaction during HTL of WBC including heating, constant-temperature reaction, and cooling. The T_C was 653 K, while the reaction time t_{rCT} for constant-temperature reaction period was 30 min. For other T_C , the curves of T vs. t_{OA} of heating period before reaching the plateau T_C are identical. For lower T_C , t_{OA} reaching T_C is shorter. In cooling stage after constant-temperature reaction region, case with lower T_C starts cooling at T_C and ends the cooling at room temperature earlier than that with higher T_C . The slopes of cooling curves are the same for cases with different T_C .

HTL products were roughly distinguished to 1) liquid containing water soluble organics (WSO), 2) crude BO, 3) crude solid product (SPC), and 4) gases. After removing liquid with WSO, acetone was poured into reactor to dissolve all available crude BO on the surface of reactor and top cover as well as SPC. The products mixing with acetone were separated into two parts, namely the extracted solid product and acetone soluble liquid containing acetone (ASL) by vacuum filtration with 47 mm fiber filter (Glass fiber filter 47 mm, Pall, Port Washington, New York). Finally, target acetone soluble BO (BO_{AS}) can be obtained by removing acetone from ASL with vacuum rotary evaporator (Vacuum rotary evaporator, N-1, Tokyo Rikakikai, Tokyo, Japan). The heating value H_{BOAS} of BO_{AS} was analyzed by the ASTM D2015 method via calorimeter (Oxygen bomb plain jacket calorimeter, model 1341, Parr Instrument, Moline, IL) [26].

2.3. Data accuracy

Duplicate measurements were conducted for some chosen tests. Standard deviations (σ_{n-1}) were presented to indicate the errors. The accuracy of data may be further revealed by the relative errors (ϵ_R) in terms of σ_{n-1} /mean value. In this work, the values of ϵ_R were 1.3–5.7% (cellulose, hemi-cellulose, and lignin) and 11.9% (alcohol–benzene solubility) for fiber compositions and 2.0–4.0% for yields of BO (Y_{BOAS}) for WBC CHTL. As for the element compositions, ϵ_R was 0.1–6.4% for C, H, and O except 16.5% for O at 653 K. The compositions of N and S were low, resulting in ϵ_R of 0.1–18.4% for N and 8.3–20% for S. Thus, except the measurements of alcohol–benzene solubility and some data of O, N, and S, fairly accurate data were ensured for this study.

3. Results and discussion

3.1. Characteristics of WBC

The WBC used in this work is the same as that employed by Chen [12]. He reported the wet-basis contents of equilibrium moisture, volatile matters, fixed carbons, and ash of WBC of 10.8, 75.5, 12.3, and 1.4 wt%, respectively, giving combustibles (M_{CB}) of 87.7 wt%. The dry-basis high heating value (H_{HD}) is 19.2 MJ kg^{-1} . The ultimate analysis of WBC performed by this study gave dry-basis compositions of C, H, O, N, S, and others (by balance) of 44.1, 6.1, 41.9, 0.2, 0.2 and 7.5 wt%, respectively. The results indicate that WBC is a moderate-quality fuel containing high contents of volatile matter and oxygen. It would need some upgrading treatments for its proper uses, such as torrefaction conducted by Chen [12] and HTL adopted by this work.

Hemi-cellulose and cellulose exhibit lower decomposition temperature than lignin [27]. Accordingly, if the major compositions of raw materials are hemi-cellulose and cellulose, yield of liquid product would be considerably significant after HTL. Consequently, the contents of hemi-cellulose, cellulose, and lignin in raw materials were assessed firstly to decide how suitable the materials can be used for manufacturing bio-oil. Table 1 shows the results of fiber analyses of WBC [12] employed by this study, comparing with those of algae of Ververis et al. [28]. In dry basis, the content of lignin of WBC is 25.5 wt% much higher than that of algae of 5.7 wt%. Moreover, hemi-cellulose of WBC of 26.8 wt% is much lower than that of algae of 61 wt% of its total mass. It can be deduced that WBC would be more difficult for HTL than algae.

3.2. Performance of HTL at various constant-temperature reaction times t_{rCT} and temperatures T_C

Corresponding to the variation of temperature T with overall reaction time t_{OA} from the beginning of reaction during HTL of WBC

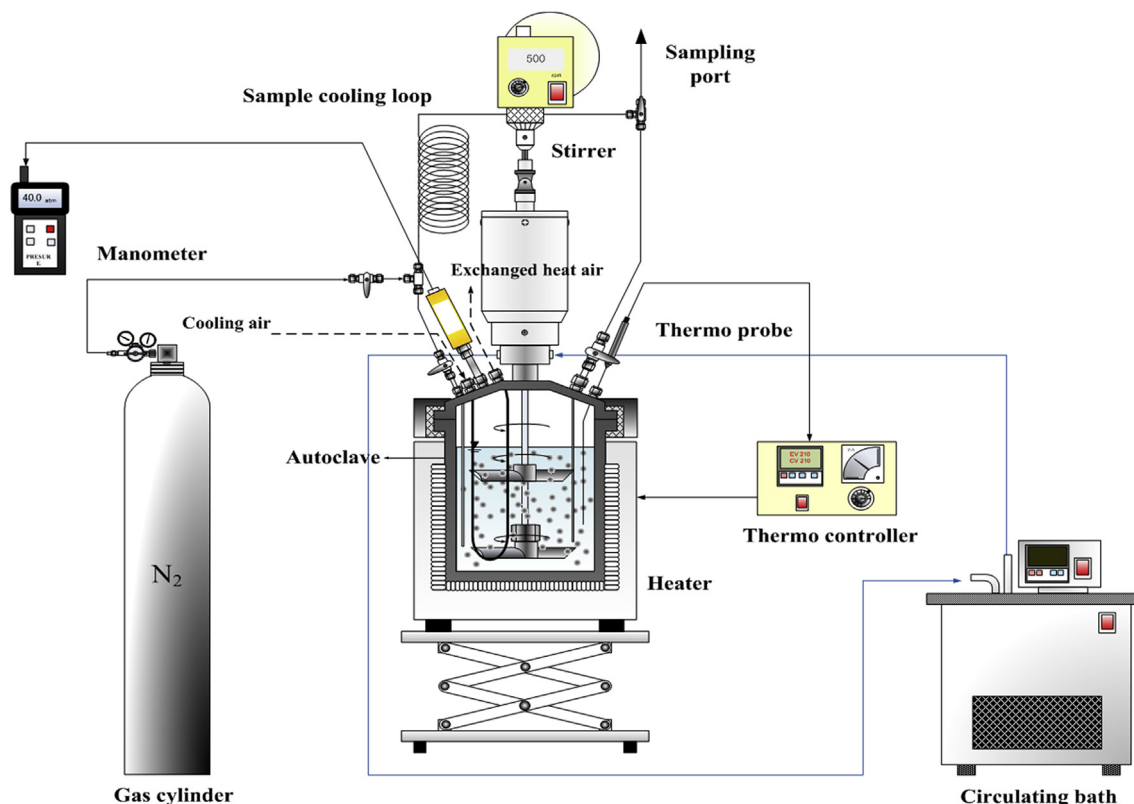


Fig. 1. Schematic diagram for experimental apparatus of hydrothermal liquefaction (HTL) and catalytic HTL (CHTL).

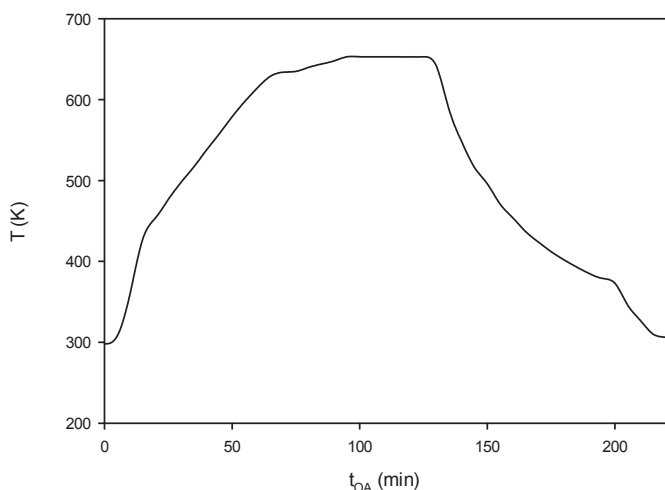


Fig. 2. Variation of temperature (T) with overall reaction time (t_{OA}) from beginning of reaction during HTL of WBC. Temperature (T_C) = 653 K (380 °C) and reaction time (t_{rCT}) = 30 min in constant-temperature reaction region.

for the case with T_C of 653 K and t_{rCT} of 30 min (reaction time for constant-temperature reaction period), the pressure variation during HTL is shown in Fig. 3. The gas pressure increases with increasing T as expected reaching about 21.4 MPa (gauge) at T_C of 653 K. Therefore, the HTL took place at very high pressure. As T decreases, pressure decreases. For other T_C , the increasing curves of pressure vs. t_{OA} during heating from room T to T_C are the same. With lower T_C , t_{OA} approaching plateau pressure corresponding to T_C is faster. For cooling period after constant-temperature reaction stage, pressure of case with lower T_C begins to decrease (as cooling

Table 1

Fiber compositions of raw algae and WBC.

Item	Raw algae	Raw WBC
Composition (dry basis in wt%)		
Cellulose	26.6 ^a	46.5 ± 2.3 ^b
Hemi-celluloses	61 ^a	26.8 ± 1.5 ^b
Lignin	5.7 ^a	25.5 ± 0.3 ^b
Alcohol–benzene solubility	—	3.3 ± 0.4 ^b
Ash	6.7 ^a	1.6

^a Computed from wet-basis data of Ververis et al. [28] with cellulose, hemi-celluloses, lignin, ash of 7.1, 16.3, 1.5, 1.8 wt%; water content = 73.3 wt%.

^b Data of Chen [12].

starts at T_C) and reduces to atmosphere pressure (as cooling ends at room temperature) quicker than that with higher T_C . For cases with different T_C , the slopes of decreasing curves of pressure vs. t_{OA} are the same. The high pressure enhances the HTL reactions not only in the liquid phase with increasing solubility of reactants but also in the gas phase which are more vigorous with higher pressure.

The yields Y_{BOAS} of bio-oil BO obtained as acetone soluble (BO_{AS}) relative to input dried WBC for HTL of WBC (denoted as WBC HTL) at various t_{rCT} and T_C are presented in Fig. 4. The values of Y_{BOAS} are low within 0.9–5.9 wt%, indicating the ineffectiveness for HTL of WBC which has high lignin content of 25.5 wt% in dry basis. Further, as shown in Fig. 2, HTL had already proceeded 90 min when reaching T_C of 653 K, however, only a very low contribution of Y_{BOAS} of 1.2 wt% was achieved at $t_{rCT} = 0$ (Fig. 4) which is the starting time of HTL at constant T_C . The corresponding results of heating values H_{BOAS} of BO_{AS} for WBC HTL are shown in Fig. 5, giving 28.1–33.5 MJ kg^{−1} higher than dry-basis high heating value H_{HD} of raw WBC of 19.2 MJ kg^{−1} and Taiwan Power Co. quality D coal standard of wet-basis high heating value (H_{HW}) of minimum

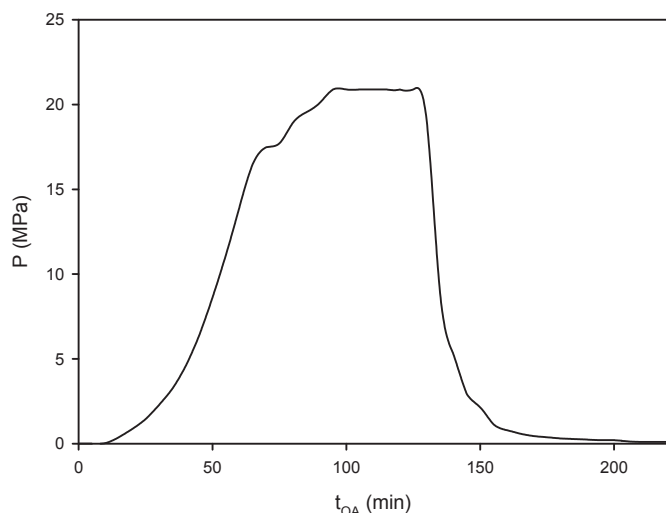


Fig. 3. Variation of pressure (P, in MPa, gauge reading) with t_{OA} during HTL of WBC. $T_C = 653$ K and $t_{rCT} = 30$ min.

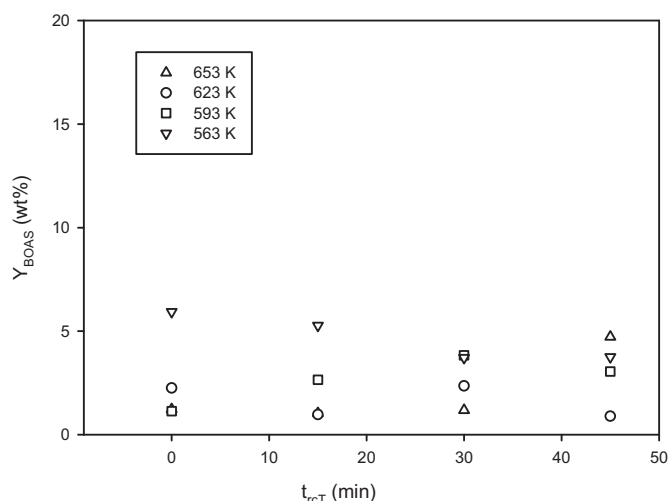


Fig. 4. Yield (Y_{BOAS}) of bio-oil obtained as acetone soluble relative to input dried WBC for HTL of WBC vs. t_{rCT} at different T_C .

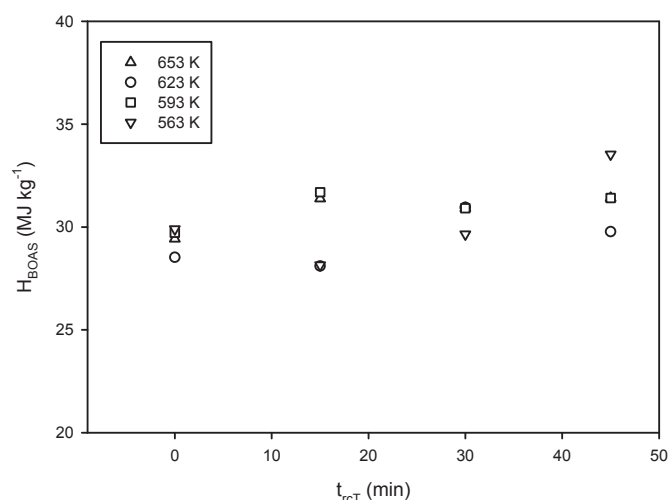


Fig. 5. Heating value (H_{BOAS}) of BO_{AS} for WBC HTL vs. t_{rCT} at different T_C .

20.9 MJ kg^{-1} [29,30]. Thus the BO_{AS} obtained can be used as liquid fuel with high heating value.

3.3. Effects of setting temperature T_C on the performance of HTL and comparison with those of catalytic HTL

HTL (WBC HTL) and CHTL using K_2CO_3 (symbolized as WBC CHTL) of WBC were conducted at different temperatures (T_C : 563, 593, 623, 653 K) with $t_{rCT} = 30$ min. In Table 2, BO_{AS} yields Y_{BOAS} of WBC HTL are all low with the highest Y_{BOAS} of 3.8 wt% at 593 K and then decreasing with further increasing temperature. The exhibition of maximum Y_{BOAS} at certain T_C for WBC HTL is consistent with the results of Sugano et al. [25] and Xu and Lancaster [9] dealing with eucalyptus and pulp/paper sludge powder, respectively, via HTL. Heating values of H_{BOAS} of BO_{AS} are shown in Table 3, which reveals that H_{BOAS} of BO_{AS} of WBC HTL at different T_C are around $29.7\text{--}31.0 \text{ MJ kg}^{-1}$, generally exhibiting an increasing trend. The above trend may be further elucidated as follows. An increasing T_C from room temperature to higher temperatures generally enhances the liquefaction reactions as expected, resulting in higher yield Y_{BOAS} of BO_{AS} with higher carbon content and thus higher H_{BOAS} . However, further increasing T_C may decompose the light components of BO_{AS} to form gaseous products, thus decreasing the Y_{BOAS} . The decrease of light components retains the dense components, also increasing the H_{BOAS} of BO_{AS} .

As for WBC CHTL, results of Tables 2 and 3 indicate that the adding of K_2CO_3 can greatly improve the Y_{BOAS} and H_{BOAS} of BO_{AS} . Although Y_{BOAS} would increase as T_C increases from room temperature to higher temperatures, it decreases with increasing temperature as T_C exceeds 563 K. The trend of variation of Y_{BOAS} for WBC CHTL with appearance of maximum value at certain T_C is similar to that for WBC HTL. However, the value of Y_{BOAS} can be raised to maximal value of 21.2 wt% at 563 K for WBC CHTL from only 3.8 wt% at 593 K for WBC HTL without K_2CO_3 catalyst. In addition, according to these results, K_2CO_3 catalyst not only enhances the Y_{BOAS} of HTL reaction of WBC but also reduces extra heat energy for reaction via the use of low T_C . The proper condition for high Y_{BOAS} is 563 K in WBC CHTL lower than 593 K in WBC HTL. Meanwhile, the highest heating value H_{BOAS} of BO_{AS} of WBC CHTL is 33.9 MJ kg^{-1} at 653 K higher than that of WBC HTL with 31.0 MJ kg^{-1} at 623 K.

According to Y_{BOAS} at each T_C , the appropriate conditions of WBC HTL and WBC CHTL can be decided. For WBC HTL, the proper T_C is 593 K with $Y_{BOAS} = 3.8 \text{ wt\%}$ and 30.9 MJ kg^{-1} . As for WBC CHTL, the suitable T_C of 563 K gives $Y_{BOAS} = 21.2 \text{ wt\%}$ and 31.6 MJ kg^{-1} .

3.4. Characteristics of bio-oil BO_{AS}

Table 4 lists the ultimate composition (dry basis in wt%) of BO_{AS} for WBC CHTL with $t_{rCT} = 30$ min and $T_C = 563\text{--}653$ K. The major contents are carbon and oxygen of 66.4–74.2 and 18.4–25.1 wt%, respectively. Thus, Y_{BOAS} is mostly contributed from matters containing carbon and oxygen, especially carbon. The contents of C exhibit a fluctuating with 66.4% at 593 K slightly lower than 68.7% at 563 K. However, despite of this fluctuation which might be attributed to the experimental error, C content generally increases with T_C . This tendency of carbonization also results in a generally increasing trend of H_{BOAS} although it also has fluctuating with 31.1 MJ kg^{-1} at 593 K lower than 31.6 MJ kg^{-1} at 563 K. H_{BOAS} is mainly contributed by C content showing the similar trend as C. Thus, an increase of T_C tends to carbonize the liquid product BO_{AS} enriching the C content along with enhancing the H_{BOAS} . At higher T_C , however, the thermal energy may also enhance the decomposition of light components of BO_{AS} to generate gaseous products as noted in previous section. This thus results in a lower Y_{BOAS} for WBC

Table 2Y_{BOAS} for HTL (WBC HTL) and CHTL (WBC CHTL) of WBC at various T_C with t_{rcT} = 30 min.

	563 K Y _{BOAS} wt%	593 K Y _{BOAS} wt%	623 K Y _{BOAS} wt%	653 K Y _{BOAS} wt%
WBC HTL	3.7	3.8	2.4	1.2
WBC CHTL	21.2 ± 0.8	15.6 ± 0.3	11.5	7.1

Table 3Heating values H_{BOAS} of BO_{AS} for WBC HTL and WBC CHTL at various T_C with t_{rcT} = 30 min.

	563 K heating value MJ kg ⁻¹	593 K heating value MJ kg ⁻¹	623 K heating value MJ kg ⁻¹	653 K heating value MJ kg ⁻¹
WBC HTL	29.7	30.9	31.0	30.9
WBC CHTL	31.6	31.1	32.0	33.9

Table 4Ultimate composition (dry basis in wt%) of BO_{AS} for WBC CHTL with t_{rcT} = 30 min and T_C = 563–653 K (290–380 °C).

	563 K	593 K	623 K	653 K
C	68.7 ± 0.9	66.4 ± 0.1	73.8 ± 1.7	74.2 ± 4.7
H	7.0 ± 0.0	6.8 ± 0.1	6.7 ± 0.1	7.4 ± 0.1
O	25.1 ± 0.1	24.1 ± 0.2	18.4 ± 0.7	22.1 ± 3.7
N	0.4 ± 0.0	0.5 ± 0.1	0.5 ± 0.1	0.5 ± 0.0
S	0.05 ± 0.01	0.08 ± 0.01	0.05 ± 0.01	0.12 ± 0.01
H/C	1.23	1.23	1.09	1.19
O/C	0.27	0.27	0.19	0.22

For WBC: C, H, O, N, and S are 44.1, 6.1, 41.9, 0.2, and 0.2 wt% in dry basis; H/C and O/C (atomic ratios) are 1.66 and 0.71.

CHTL at higher T_C. Comparing with the contents of C, H, O, N, and S of raw WBC of 44.1, 6.1, 41.9, 0.2, and 0.2 wt%, those of C (66.4–74.2), H (6.7–7.4), and N (0.4–0.5) of BO_{AS} for WBC CHTL increase, while those of O (18.4–25.1) and S (0.05–0.12) decrease. Hence, CHTL concentrates the contents of C and H especially C in BO_{AS}. The higher carbon content of BO_{AS} for WBC CHTL than that of raw WBC also ensures the higher heating value H_{BOAS} of BO_{AS} for WBC CHTL. Moreover, the atomic ratios of H/C (1.09–1.23) and O/C (0.19–0.27) of BO_{AS} for WBC CHTL are lower than those of WBC of 1.66 and 0.71, respectively. Thus, the quality of BO_{AS} is better than that of raw WBC. It is worth noting that the relative improvement of H_{BOAS} from WBC HTL to WBC CHTL is low to moderate of 0.5–9.5%, while that of Y_{BOAS} is high of 473–500%. The Y_{BOAS} counts the amount of product BO_{AS} formed due to the liquefaction of solid WBC. The alkaline CHTL significantly enhances the conversion of carbonaceous WBC to BO_{AS} comparing with sole HTL which is not effective especially to high lignin-containing WBC. The BO_{AS} formed via HTL and CHTL may be different in constituents as reflected by their difference of H_{BOAS}, however, both containing essentially C, H, and O. Moreover, the contents of C and H of BO_{AS} for HTL (e.g., C = 65.6–68.4 wt% and H = 5.5–5.6 wt% at 583–613 K) though are lower than those for CHTL (e.g., C = 66.4–74.2 wt% and H = 6.7–7.4 wt% at 563–653 K) but not much, thus resulting in difference of H_{BOAS} lesser than that of Y_{BOAS}. A verification of this with a statistical comparative analysis on Y_{BOAS} and H_{BOAS} would be helpful to further justify these points.

Furthermore, from the carbon content and Y_{BOAS}, the yield (Y_{BOCC}) of organic C of BO_{AS} relative to C in input dry-ash-free WBC for WBC CHTL can be computed. This gives Y_{BOCC} of 33.1, 23.6, 19.2, and 11.9 wt% at T_C of 563, 593, 623, and 653 K, respectively. The results clearly indicate that condition at 563 K gives higher yield of BO_{AS} than other said setting temperatures. The temperature variation trend of Y_{BOCC} exhibiting maximum at 563 K is consistent with that of Y_{BOAS}. Thus, a higher Y_{BOAS} results in a higher Y_{BOCC}, i.e., the yield of organic C, as reflected from the higher C content in BO_{AS}.

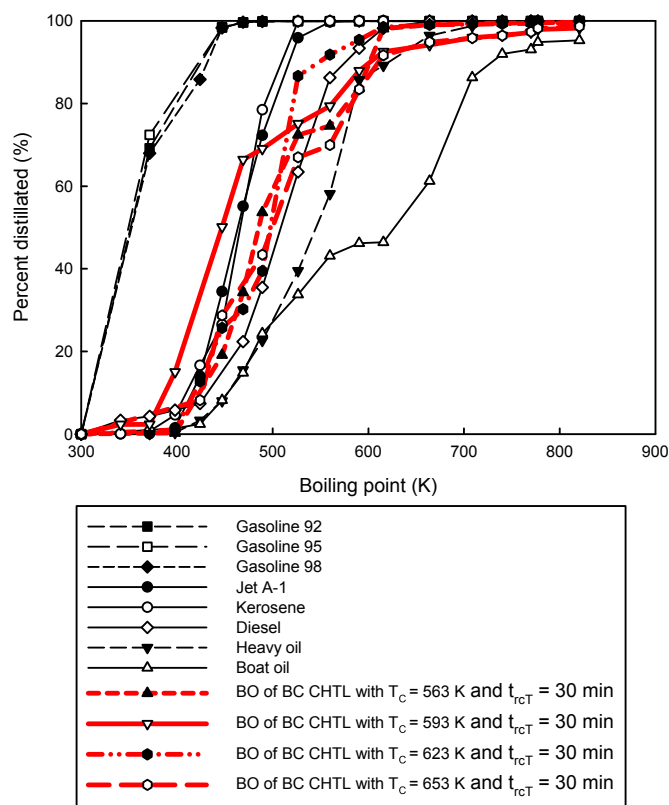


Fig. 6. Simulated distillation characteristic of BO_{AS} (i.e., BO) for WBC CHTL (i.e., BC CHTL) with T_C = 563–653 K (290–380 °C) and t_{rcT} = 30 min comparing with those of different fuels.

Fig. 6 presents the simulated distillation characteristic (SDC) of BO_{AS} for WBC CHTL with T_C = 563–593 K (290–320 °C) and t_{rcT} = 30 min comparing with those of different fuels. The results reveal that SDC of BO_{AS} is close to diesel, while better than heavy oil and boat oil. The atomic H/C ratios of 1.09–1.23 between 1 (such as $-\text{CH}=\text{CH}-$) and 2 (such as $-\text{CH}_2-\text{CH}_2-$) indicate that BO_{AS} obtained contains olefins constituents. It can be refined to produce diesel or directly used in boiler industry and ship transportation.

4. Conclusions

- As T_C of CHTL increases, H_{BOAS} of BO_{AS} generally increases but Y_{BOAS} exhibits a convex trend.
- The addition of K₂CO₃ for CHTL can improve Y_{BOAS} and H_{BOAS} of BO_{AS}.

3. The effects of K_2CO_3 on enhancing Y_{BOAS} and H_{BOAS} of BO_{AS} for CHTL are pronounced for WBC containing significant lignin.
4. Based on the highest Y_{BOAS} , the appropriate T_C for WBC HTL and WBC CHTL are 593 and 563 K, respectively.
5. The SDC of BO_{AS} is close to diesel, whereas it is superior to heavy oil and boat oil.

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Nomenclature

AS	acetone soluble
ASL	acetone soluble liquid containing acetone
BO_{AS}	bio-oil BO obtained as AS
CHTL	catalytic HTL using K_2CO_3
H_{BOAS}	heating value of bio-oil BO obtained as AS, $MJ\ kg^{-1}$
H_{HD}	dry-basis high heating value of solid, $MJ\ kg^{-1}$
H_{HW}	wet-basis high heating value, $MJ\ kg^{-1}$
M_L	lignin content of biomass, wt%
SPC	crude solid product
T	temperature, K or $^{\circ}C$
T_C	setting temperature for HTL, K or $^{\circ}C$
t	time, min
t_{OA}	overall reaction time from beginning of reaction during HTL of WBC consisting heating, constant-temperature reaction, and cooling, min.
t_{rCT}	reaction time for constant-temperature reaction period (i.e., at constant T_C), min
Y_{BOAS}	yield of bio-oil BO obtained as AS relative to mass of input dried WBC, wt%
Y_{BOCC}	yield of organic C of BO_{AS} relative to C in input dry-ash-free WBC, wt%
ε_R	relative error in term of σ_{n-1} /mean value
σ_{n-1}	standard deviation

Abbreviation

BO	bio-oil
daf	dry ash free
HTL	hydrothermal liquefaction
SDC	simulated distillation characteristic
WBC	waste bamboo chopsticks
WSO	liquid containing water soluble organics

References

- [1] IEA. Bioenergy. Paris, France: International Energy Agency; 2015.
- [2] Bacovsky D. 2nd Generation biofuels – state of the art: an overview on demonstration plants around the world. A3PS 2009 Conference on Alternative Propulsion Systems and Energy Carriers; 2009 Oct 16; Vienna, Austria.
- [3] Hu G, Heitmann JA, Rojaas OJ. Feedstock pretreatment strategies for producing ethanol from wood, bark, and forest residues. *BioResources* 2008;3:270–94.
- [4] Hung ZS, Chang CC, Chang CFH, Lin YS, Ji DR, Chang CY, et al. Autoclaving treatment of wasted disposable bamboo chopsticks. *J Taiwan Inst Chem Eng* 2013;44:1010–5.
- [5] Kaar WE, Holtzaple MT. Using lime pretreatment to facilitate the enzymic hydrolysis of corn stover. *Biomass Bioenerg* 2000;18:189–99.
- [6] Laureano-Perez L, Teymouri F, Alizadeh H, Dale BE. Understanding factors that limit enzymatic hydrolysis of biomass. *Appl Biochem Biotechnol* 2005;121:1081–99.
- [7] Pimchui A, Dutta A, Basu P. Torrefaction of agriculture residue to enhance combustible properties. *Energy Fuel* 2010;24:4638–45.
- [8] Refaat AA. Biofuels from waste materials. In: Sayigh A, van Sark WGJHM, Kaldellis JK, Kalogirou SA, Cruden AJ, Roddy DJ, et al., editors. *Comprehensive Renewable Energy*. Oxford, UK: Elsevier; 2012.
- [9] Xu CB, Lancaster J. Conversion of secondary pulp/paper sludge powder to liquid oil products for energy recovery by direct liquefaction in hot-compressed water. *Water Res* 2008;42:1571–82.
- [10] Zhu Z, Toor SS, Rosendahl L, Yu DH, Chen GY. Influence of alkali catalyst on product yield and properties via hydrothermal liquefaction of barley straw. *Energy* 2015;80:284–92.
- [11] Asada C, Kita A, Sasaki C, Nakamura Y. Ethanol production from disposable aspen chopsticks using delignification pretreatments. *Carbohydr Polym* 2011;85:196–200.
- [12] Chen YH. The Torrefaction of Waste Bamboo Chopsticks to Manufacturing Solid Biofuel [Master Thesis]. Taipei (Taiwan): National Taiwan Univ.; 2016.
- [13] Chiang KY, Chen YS, Tsai WS, Lu CH, Chien KL. Effect of calcium based catalyst on production of synthesis gas in gasification of waste bamboo chopsticks. *Int J Hydrogen Energy* 2012;37:13737–45.
- [14] Chiang KY, Chien KL, Lu CH. Hydrogen energy production from disposable chopsticks by a low temperature catalytic gasification. *Int J Hydrogen Energy* 2012;37:15672–80.
- [15] GC. China and Japan contribute about 90% of annual consumption of 80 billion pairs of wood chopsticks. *International Environmental News*. Hong Kong: Green Council; 2008.
- [16] Akhtar J, Amin NAS. A review on process conditions for optimum bio-oil yield in hydrothermal liquefaction of biomass. *Renew Sust Energ Rev* 2011;15:1615–24.
- [17] Christensen PR, Morup AJ, Mamakhel A, Glasius M, Becker J, Iversen BB. Effects of heterogeneous catalyst in hydrothermal liquefaction of dried distillers grains with solubles. *Fuel* 2014;123:158–66.
- [18] Duan PG, Savage PE. Hydrothermal liquefaction of a microalga with heterogeneous catalysts. *Ind Eng Chem Res* 2011;50:52–61.
- [19] Goyal HB, Seal D, Saxena RC. Bio-fuels from thermochemical conversion of renewable resources: a review. *Renew Sust Energ Rev* 2008;12:504–17.
- [20] Barreiro DL, Prins W, Ronsse F, Brilman W. Hydrothermal liquefaction (HTL) of microalgae for biofuel production: state of the art review and future prospects. *Biomass Bioenerg* 2013;53:113–27.
- [21] Tekin K, Karagöz S, Bektaş S. Effect of sodium perborate monohydrate concentrations on product distributions from the hydrothermal liquefaction of Scotch pine wood. *Fuel Process Technol* 2013;110:17–23.
- [22] Toor SS, Rosendahl L, Rudolf A. Hydrothermal liquefaction of biomass: a review of subcritical water technologies. *Energy* 2011;36:2328–42.
- [23] Yuan ZS, Cheng SN, Leitch M, Xu CB. Hydrolytic degradation of alkaline lignin in hot-compressed water and ethanol. *Bioresour Technol* 2010;101:9308–13.
- [24] Brand S, Hardi F, Kim J, Suh DJ. Effect of heating rate on biomass liquefaction: differences between subcritical water and supercritical ethanol. *Energy* 2014;68:420–7.
- [25] Sugano M, Takagi H, Hirano K, Mashimo K. Hydrothermal liquefaction of plantation biomass with two kinds of wastewater from paper industry. *J Mater Sci* 2008;43:2476–86.
- [26] ASTM. Standard Test Method for Gross Calorific Value of Coal and Coke by the Adiabatic Bomb Calorimeter. West Conshohocken, PA: American Society for Testing and Materials; 2000.
- [27] BSML. Methods of CNS (Chinese National Standards) for fiber analyses, alcohol-benzene solubility [CNS 4713-2005], holocellulose [CNS 6948-1981], alpha-cellulose [CNS 10865-1984], hemicellulose, acid-insoluble lignin [CNS-2721-2010], and acid-soluble lignin [CNS 12108-1987]. Taipei, Taiwan: Bureau of Standards, Metrology & Inspection; 2015.
- [28] Ververis C, Georgiou K, Danielidis D, Hatzinikolaou DG, Santas P, Santas R, et al. Cellulose, hemicelluloses, lignin and ash content of some organic materials and their suitability for use as paper pulp supplements. *Bioresour Technol* 2007;98:296–301.
- [29] Jian JL, Yu MJ. The verification report of Indonesia coal. Taipei, Taiwan: Taiwan Power Company; 2010 (in Chinese).
- [30] TPC. Coal standard of Taiwan Power Company. Taipei, Taiwan: Taiwan Power Company; 2010.